

Project name : “Individual_Assignment_3”

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Course: HDS 5310

Project Purpose: This analysis examine the relationships between the ease of voting variable and sex, marital status, political party, and employment status.

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```
# Load necessary libraries
```

```
library(haven)
library(descr)
library(lsr)
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4    ✓ readr      2.1.5
## ✓ forcats    1.0.0    ✓ stringr   1.5.1
## ✓ ggplot2    3.5.1    ✓ tibble    3.2.1
## ✓ lubridate  1.9.4    ✓ tidyr     1.3.1
## ✓ purrr      1.0.2
## — Conflicts — tidyverse_conflicts() —
## * dplyr::filter() masks stats::filter()
## * dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

STEP 1 : Read the dataset

```
vote <- read_sav("C:/Users/drhud/Downloads/Individual Assignment 3 (1)/Individual Assignment 3/pew_apr_19-23_2017_weekly_ch5.sav")
```

STEP 2 : Data cleaning

```
vote.cleaned <- vote %>%
  select(pew1a, pew1b, race, sex, mstatus, ownhome, employ, polparty) %>% #selecting relevant variables
  zap_labels() %>% # Removing long labels
  mutate(pew1a = recode_factor(.x = pew1a, # Recoding values for ease of understanding
    `1` = 'Register to vote',
    `2` = 'Make easy to vote',
    `5` = NA_character_,
    `9` = NA_character_)) %>%
  rename(easy.vote = pew1a) %>% # Renaming column
  mutate(pew1b = recode_factor(.x = pew1b,
    `1` = 'Require to vote',
    `2` = 'Choose to vote',
    `5` = NA_character_,
    `9` = NA_character_)) %>%
  rename(require.vote = pew1b) %>%
  mutate(race = recode_factor(.x = race,
    `1` = 'White non-Hispanic',
    `2` = 'Black non-Hispanic',
    `3` = 'Hispanic',
    `4` = 'Hispanic',
    `5` = 'Hispanic',
    `6` = 'Other',
    `7` = 'Other',
    `8` = 'Other',
    `9` = 'Other',
    `10` = 'Other',
    `99` = NA_character_)) %>%
  mutate(sex = recode_factor(.x = sex,
    `1` = 'Male',
    `2` = 'Female')) %>%
  mutate(ownhome = recode_factor(.x = ownhome,
    `1` = 'Owned',
    `2` = 'Rented',
```

```
`8` = NA_character_,  
`9` = NA_character_))
```

```
# Display summary of cleaned dataset  
summary(vote.cleaned)
```

```
##           easy.vote           require.vote           race  
## Register to vote :408 Require to vote:205 White non-Hispanic:646  
## Make easy to vote:593 Choose to vote :806 Black non-Hispanic:129  
## NA's             : 27 NA's             : 17 Hispanic           :150  
##                                                         Other           : 78  
##                                                         NA's             : 25  
##  
##           sex           mstatus           ownhome           employ           polparty  
## Male :495 Min. :1.000 Owned :678 Min. :1.000 Min. :0.000  
## Female:533 1st Qu.:2.000 Rented:328 1st Qu.:1.000 1st Qu.:1.000  
##           Median :3.000 NA's : 22 Median :2.000 Median :2.000  
##           Mean :3.219 Mean :2.527 Mean :2.475  
##           3rd Qu.:5.000 3rd Qu.:3.000 3rd Qu.:3.000  
##           Max. :9.000 Max. :9.000 Max. :9.000
```

```
# Additional Cleaning for Marital Status, Employment, and Political Party  
vote.cleaned <- vote.cleaned %>%  
  mutate(mstatus = recode_factor(.x = mstatus,  
    `1` = 'Single, that is never married',  
    `2` = 'Single, living with a partner',  
    `3` = 'Married',  
    `4` = 'Separated',  
    `5` = 'Widowed',  
    `6` = 'Divorced',  
    `9` = NA_character_)) %>%  
  mutate(employ = recode_factor(.x = employ,  
    `1` = 'Employed full-time',  
    `2` = 'Employed part-time',  
    `3` = 'Not employed',  
    `4` = 'Not employed',  
    `5` = 'Not employed',  
    `6` = 'Not employed',  
    `7` = 'Not employed',  
    `8` = 'Not employed',  
    `9` = 'Refused')) %>%  
  mutate(polparty = recode_factor(.x = polparty,  
    `1` = 'A Republican',  
    `2` = 'A Democrat',  
    `3` = 'An Independent',  
    `0` = NA_character_,  
    `8` = NA_character_,  
    `9` = NA_character_))
```

```
# Display summary after further cleaning  
summary(vote.cleaned)
```

```
##           easy.vote           require.vote           race  
## Register to vote :408 Require to vote:205 White non-Hispanic:646  
## Make easy to vote:593 Choose to vote :806 Black non-Hispanic:129  
## NA's             : 27 NA's             : 17 Hispanic           :150  
##                                                         Other           : 78  
##                                                         NA's             : 25  
##  
##           sex           mstatus           ownhome  
## Male :495 Single, that is never married:229 Owned :678  
## Female:533 Single, living with a partner: 79 Rented:328  
##           Married           :422 NA's : 22  
##           Separated           : 21  
##           Widowed           :126  
##           Divorced           :139  
##           NA's             : 12  
##           employ           polparty  
## Employed full-time:414 A Republican :249  
## Employed part-time:133 A Democrat   :314  
## Not employed :478 An Independent:398  
## Refused : 3 NA's : 67  
##  
##
```

STEP 3 : Descriptive statistics

#STEP 3a : Descriptive statistics to examine the relationships between the ease of voting variable and sex

```
prop.table(x = table(sex = vote.cleaned$sex, Voting.ease = vote.cleaned$easy.vote))
```

```
##           Voting.ease
## sex      Register to vote Make easy to vote
## Male           0.1988012           0.2787213
## Female          0.2087912           0.3136863
```

#Interpretation The descriptive statistics show that both males (27.87%) and females (31.37%) are more likely to support making voting easier rather than requiring registration. While females show slightly higher support for easing voting compared to males, the overall distribution suggests that opinions on this issue are relatively similar across genders.

#STEP 3b : Descriptive statistics to examine the relationships between the ease of voting variable and marital status

```
prop.table(x = table(mstatus = vote.cleaned$mstatus,
                     Voting.ease = vote.cleaned$easy.vote))
```

```
##           Voting.ease
## mstatus      Register to vote Make easy to vote
## Single, that is never married  0.074747475      0.149494949
## Single, living with a partner  0.030303030      0.047474747
## Married                        0.188888889      0.230303030
## Separated                      0.006060606      0.014141414
## Widowed                       0.061616162      0.064646465
## Divorced                      0.046464646      0.085858586
```

#Interpretation The descriptive statistics shows that married individuals are the most likely to support both registering to vote (18.89%) and making voting easier (23.03%) compared to other groups. Single individuals (never married) also show moderate support for making voting easier (14.95%), though their support for registration is lower (7.47%). Divorced (8.59%) and widowed individuals (6.46%) lean slightly more towards making voting easier rather than requiring registration. Those living with a partner and separated individuals show the lowest support for either category. Overall, marital status appears to influence voting ease preferences, with married individuals being the most engaged in both aspects.

#STEP 3a=c : Descriptive statistics to examine the relationships between the ease of voting variable and employment status

```
prop.table(x = table(employ = vote.cleaned$employ,
                     Voting.ease = vote.cleaned$easy.vote))
```

```
##           Voting.ease
## employ      Register to vote Make easy to vote
## Employed full-time  0.160839161      0.245754246
## Employed part-time  0.049950050      0.076923077
## Not employed        0.195804196      0.267732268
## Refused             0.000999001      0.001998002
```

#Interpretation The descriptive statistics shows that unemployed individuals have the highest support for both registering to vote (19.58%) and making voting easier (26.77%). Full-time employees also show notable support, with 16.08% favoring registration and 24.57% supporting easier voting. Part-time employees, however, express the lowest support for both registering (4.99%) and making voting easier (7.69%). Overall, employment status seems to influence opinions on voting ease, with unemployed individuals showing the highest engagement in both aspects, while part-time workers are the least likely to support either measure.

#STEP 3d : Descriptive statistics to examine the relationships between the ease of voting variable and political party

```
prop.table(x = table(polparty = vote.cleaned$polparty,
                     Voting.ease = vote.cleaned$easy.vote))
```

```
##           Voting.ease
## polparty      Register to vote Make easy to vote
## A Republican  0.16719915           0.09158679
## A Democrat    0.05963791           0.26837061
## An Independent 0.17784878           0.23535676
```

#Interpretation The descriptive statistics shows that Republicans are more likely to support registering to vote (16.72%) but have the lowest support for making voting easier (9.16%). Democrats, on the other hand, strongly favor making voting easier (26.84%), while their support for voter registration is much lower (5.96%). Independents fall in between, with 17.78% supporting registration and 23.53% favoring making voting easier. This suggests that Republicans are more inclined toward voter registration requirements, while Democrats strongly support easing voting access. Independents lean toward making voting easier but still show considerable support for registration.

STEP 4 : NHST process and test whether or not there is a relationship between opinion on ease of voting and sex

#NHST Step 1: Write the null and alternate hypotheses H_0 : There is no relationship between opinion on ease of voting opinion and sex. H_A : There is a relationship between opinion on ease of voting opinion and sex.

#Remove missing values for hypothesis testing

```
vote.cleaned <- vote.cleaned %>%
  drop_na()
```

#NHST Step 2: Get the test statistic

```
chisq.test(x = vote.cleaned$easy.vote,
  y = vote.cleaned$sex)
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: vote.cleaned$easy.vote and vote.cleaned$sex
## X-squared = 0.058059, df = 1, p-value = 0.8096
```

The test statistic is χ^2 : 0.058059

#NHST Step 3: Calculate the probability that your test statistic is at least as big as it is if there is no relationship (i.e., the null is true) The probability that the null hypothesis is true is 0.8096

#NHST Step 4: If the probability that the null is true is very small, usually less than 5%, reject the null hypothesis. $p = 0.8096$, which is much greater than 0.05. Since $p > 0.05$, we fail to reject the null hypothesis (H_0).

#NHST Step 5: If the probability that the null is true is not small, usually 5% or greater, retain the null hypothesis Since $p = 0.8096 (> 0.05)$, we retain the null hypothesis. This means that there is no statistically significant relationship between opinion on ease of voting and sex.

#Final interpretation We used the chi-squared test to test the null hypothesis that there was no relationship between opinions on ease of voting and sex. The test result [$\chi^2 = 0.058059$; $p = 0.8096$] indicates a high p-value, meaning the observed differences are likely due to chance. Since $p = 0.8096 (> 0.05)$, we fail to reject the null hypothesis, suggesting that there is no statistically significant association between views on ease of voting and sex. This means that people's opinions on ease of voting do not differ significantly based on their sex.

#NHST Step 1: Write the null and alternate hypotheses H_0 : There is no relationship between opinion on ease of voting and marital status. H_A : There is a relationship between opinion on ease of voting and marital status.

#NHST Step 2: Get the test statistic

```
chisq.test(x = vote.cleaned$easy.vote,
  y = vote.cleaned$mstatus)
```

```
##
## Pearson's Chi-squared test
##
## data: vote.cleaned$easy.vote and vote.cleaned$mstatus
## X-squared = 14.11, df = 5, p-value = 0.01493
```

The test statistic is χ^2 : 14.11

#NHST Step 3: Calculate the probability that your test statistic is at least as big as it is if there is no relationship (i.e., the null is true) The probability that the null hypothesis is true is 0.01493.

#NHST Step 4: If the probability that the null is true is very small, usually less than 5%, reject the null hypothesis. $p = 0.01493$, which is less than 0.05. Since $p < 0.05$, we reject the null hypothesis (H_0).

#NHST Step 5: If the probability that the null is true is not small, usually 5% or greater, retain the null hypothesis Not applicable here, as $p < 0.05$ ($p = 0.01493$). This means that there is a statistically significant relationship between opinion on ease of voting and marital status.

#Final interpretation We used the chi-squared test to test the null hypothesis that there was no relationship between opinions on ease of voting and marital status. The test result [$\chi^2 = 14.11$; $p = 0.01493$] indicates a low p-value, meaning the observed differences are unlikely to be due to chance. Since $p = 0.01493 (< 0.05)$, we reject the null hypothesis, suggesting that there is a statistically significant association between views on ease of voting and marital status. This means that people's opinions on ease of voting differ significantly based on their marital status.

#NHST Step 1: Write the null and alternate hypotheses H_0 : There is no relationship between opinion on ease of voting and employment status. H_A : There is a relationship between opinion on ease of voting and employment status.

#NHST Step 2: Get the test statistic & p-value

```
chisq.test(x = vote.cleaned$easy.vote,
  y = vote.cleaned$employ)
```

```
##
## Pearson's Chi-squared test
##
## data: vote.cleaned$easy.vote and vote.cleaned$employ
## X-squared = 1.6976, df = 2, p-value = 0.4279
```

The test statistic is χ^2 : 1.6976

#NHST Step 3: Calculate the probability that your test statistic is at least as big as it is if there is no relationship (i.e., the null is true) The probability that the null hypothesis is true is 0.4279.

#NHST Step 4: If the probability that the null is true is very small, usually less than 5%, reject the null hypothesis. $p = 0.4279$, which is much greater than 0.05. Since $p > 0.05$, we fail to reject the null hypothesis (H_0).

#NHST Step 5: If the probability that the null is true is not small, usually 5% or greater, retain the null hypothesis Since $p = 0.4279 (> 0.05)$, we retain the null hypothesis. This means that there is no statistically significant relationship between opinion on ease of voting and employment status.

#Final interpretation We used the chi-squared test to test the null hypothesis that there was no relationship between opinions on ease of voting and employment status. The test result [$\chi^2 = 1.6976$; $p = 0.4279$] indicates a high p-value, meaning the observed differences are likely due to chance. Since $p = 0.4279 (> 0.05)$, we fail to reject the null hypothesis, suggesting that there is no statistically significant association between views on ease of voting and employment status. This means that people's opinions on ease of voting do not differ significantly based on their employment status.

#NHST Step 1: Write the null and alternate hypotheses H_0 : There is no relationship between opinion on ease of voting and political party . H_A : There is a relationship between opinion on ease of voting and political party

#NHST Step 2: Get the test statistic

```
chisq.test(x = vote.cleaned$easy.vote,
           y = vote.cleaned$polparty)
```

```
##
## Pearson's Chi-squared test
##
## data: vote.cleaned$easy.vote and vote.cleaned$polparty
## X-squared = 113.03, df = 2, p-value < 2.2e-16
```

The test statistic is χ^2 : 113.03

#NHST Step 3: Calculate the probability that your test statistic is at least as big as it is if there is no relationship (i.e., the null is true) The probability that the null hypothesis is true is $p < 2.2e-16$

#NHST Step 4: If the probability that the null is true is very small, usually less than 5%, reject the null hypothesis. $p < 2.2e-16$, which is much smaller than 0.05. Since $p < 0.05$, we reject the null hypothesis (H_0).

#NHST Step 5: If the probability that the null is true is not small, usually 5% or greater, retain the null hypothesis Not applicable here, as $p < 0.05$. This means there is a statistically significant relationship between opinion on ease of voting and political party affiliation.

#Final interpretation We used the chi-squared test to test the null hypothesis that there was no relationship between opinions on ease of voting and political party affiliation. The test result [$\chi^2 = 113.03$; $p < 2.2e-16$] indicates an extremely low p-value, meaning the observed differences are highly unlikely to be due to chance. Since $p < 0.05$, we reject the null hypothesis, suggesting that there is a statistically significant association between views on ease of voting and political party affiliation. This means that people's opinions on ease of voting differ significantly based on their political party affiliation.

Step 5 :Using standardized residuals following chi-squared tests

```
# chi-squared examining ease of voting and marital status
CrossTable(x = vote.cleaned$easy.vote,
           y = vote.cleaned$mstatus,
           expected = TRUE, # expected value
           prop.c = FALSE, # default to include proportions in the column (disabling that by assigning FALSE)
           prop.t = FALSE, # default to include proportion out of total (disabling that by assigning FALSE)
           prop.chisq = FALSE, # default to include proportions of the chi-squared (disabling that by assigning FALSE)
           chisq = TRUE, # chi-squared
           sresid = TRUE) #standardized residuals
```

```
##      Cell Contents
## |-----|
## |              N |
## |      Expected N |
## |      N / Row Total |
## |      Std Residual |
## |-----|
##
## =====
##              vote.cleaned$mstatus
## vt.cl$.      S, t i      S, l w      Married      Separatd      Widowed      Divorcd      Total
## -----
## Rgs t v          63          25          167          6          57          43          361
##              78.4          27.2          151.5          8.1          46.7          49.1
##              0.175          0.069          0.463          0.017          0.158          0.119          0.406
##              -1.736          -0.423          1.262          -0.744          1.507          -0.875
## -----
## M e t v          130          42          206          14          58          78          528
##              114.6          39.8          221.5          11.9          68.3          71.9
##              0.246          0.080          0.390          0.027          0.110          0.148          0.594
##              1.436          0.350          -1.044          0.616          -1.246          0.724
## -----
## Total          193          67          373          20          115          121          889
## =====
##
## Statistics for All Table Factors
##
## Pearson's Chi-squared test
```

```
## -----
## Chi^2 = 14.10961      d.f. = 5      p = 0.0149
```

Values of the standardized residuals that are higher than 1.96 or lower than -1.96 indicate that the observed value in that group is much higher or lower than the expected value. Based on the standardized residuals: -> Fewer single individuals (never married) than expected supported the idea of "Register to vote" (std. res. = -1.736), meaning that this group was less likely to believe that voter registration should be required. -> More married individuals than expected supported "Register to vote" (std. res. = 1.262), suggesting that this group was more likely to believe that people should register to vote. -> More widowed individuals than expected supported "Register to vote" (std. res. = 1.507), indicating a stronger preference for voter registration requirements. -> Fewer widowed individuals than expected supported "Make easy to vote" (std. res. = -1.246), suggesting that this group was less likely to favor making voting easier. -> Fewer married individuals than expected supported "Make easy to vote" (std. res. = -1.044), indicating that this group was less likely to prefer reducing barriers to voting.

#Final Interpretation We used the chi-squared test to test the null hypothesis that there was no relationship between opinions on ease of voting and marital status. The test result [$\chi^2 = 14.11$; $p = 0.0149$] indicates a statistically significant relationship, meaning that people's views on ease of voting differ significantly based on marital status. The standardized residual analysis suggests that widowed and married individuals were more likely to support voter registration requirements, while single (never married) and widowed individuals were less likely to support making voting easier. These patterns indicate that opinions on ease of voting vary across marital status groups in a way that is unlikely due to chance.

```
# chi-squared examining ease of voting and political party
CrossTable(x = vote.cleaned$easy.vote,
           y = vote.cleaned$polparty,
           expected = TRUE, # expected value
           prop.c = FALSE, # default to include proportions in the column (disabling that by assigning FALSE)
           prop.t = FALSE, # default to include proportion out of total (disabling that by assigning FALSE)
           prop.chisq = FALSE, # default to include proportions of the chi-squared (disabling that by assigning FALSE)
           chisq = TRUE, # chi-squared
           sresid = TRUE) #standardized residuals
```

```
##      Cell Contents
## |-----|
## |              N |
## |      Expected N |
## |      N / Row Total |
## |      Std Residual |
## |-----|
##
## =====
##                      vote.cleaned$polparty
## vote.cleaned$easy.vote  A Republican  A Democrat  An Independent  Total
## -----
## Register to vote              149           54           158         361
##                               94.6         118.6         147.8
##                               0.413         0.150         0.438      0.406
##                               5.591        -5.930         0.838
## -----
## Make easy to vote              84          238          206         528
##                               138.4        173.4         216.2
##                               0.159         0.451         0.390      0.594
##                               -4.623         4.903        -0.693
## -----
## Total                233           292           364         889
## =====
##
## Statistics for All Table Factors
##
## Pearson's Chi-squared test
## -----
## Chi^2 = 113.0252      d.f. = 2      p <2e-16
```

#Interpretation of Standardized Residuals -> More Republicans than expected supported "Register to vote" (std. res. = 5.591), while fewer Democrats than expected supported it (std. res. = -5.930). -> Fewer Republicans than expected supported "Make easy to vote" (std. res. = -4.623), while more Democrats than expected supported it (std. res. = 4.903). -> Independents do not show a strong deviation from expected values, since both residuals are between -1.96 and 1.96. # Final Interpretation We used the chi-squared test to examine the relationship between opinions on ease of voting and political party affiliation. The test result [$\chi^2 = 113.03$; $p < 2e-16$] indicates a statistically significant association, meaning views on ease of voting vary significantly by political party. The standardized residuals suggest that Republicans are more likely to support voter registration requirements, while Democrats are more likely to support making voting easier. Independents did not show strong deviations in either category.

STEP 6 - Computing the Cramér's V statistic (ease.to.vote and marital status)

```
cramersV(x = vote.cleaned$easy.vote, y = vote.cleaned$mstatus)
```

```
## [1] 0.1259815
```

#Interpretation * Small or weak effect size for $V = .1$ * Medium or moderate effect size for $V = .3$ * Large or strong effect size for $V = .5$ In this case, the effect size is small.

#Final Interpretation There is a statistically significant relationship between opinions on ease of voting and marital status, but the effect size is small. The Cramér's V value ($V = 0.126$) falls within the weak effect size range, indicating that while marital status is associated with opinions on ease of voting, the strength of this relationship is weak.

#Computing the Cramér's V statistic (ease.to.vote and polparty)

```
cramersV(x = vote.cleaned$easy.vote, y = vote.cleaned$polparty)
```

```
## [1] 0.3565634
```

#Interpretation * Small or weak effect size for $V = .1$ * Medium or moderate effect size for $V = .3$ * Large or strong effect size for $V = .5$ In this case, the effect size is medium.

#Final Interpretation There is a statistically significant relationship between opinions on ease of voting and political party affiliation, with a moderate to strong effect size. The Cramér's V value ($V = 0.357$) falls within the moderate effect size range, indicating that while political party affiliation is associated with opinions on ease of voting, the strength of this relationship is moderate .

Step 7 : Checking the assumptions for employ and sex variables with easy.vote

#Violation 1: Violating the assumption that the variables must be nominal or ordinal -> Ease of Voting (easy.vote) is nominal (categories: "Register to vote", "Make easy to vote"). -> Sex (sex) is nominal (categories: "Male", "Female"). -> Employment Status (employ) is ordinal (categories: "Employed full-time", "Employed part-time", "Not employed"). Since both variables are categorical, the chi-square test is appropriate. This assumption is satisfied, we proceed with further checks.

#Violation 2: Violating the assumption of expected values of 5 or higher in at least 80% of groups

```
#checking expected values for ease of vote and employment status
```

```
CrossTable(x = vote.cleaned$easy.vote,
  y = vote.cleaned$employ,
  expected = TRUE, # expected value
  prop.c = FALSE, # default to include proportions in the column (disabling that by assigning FALSE)
  prop.t = FALSE, # default to include proportion out of total (disabling that by assigning FALSE)
  prop.chisq = FALSE, # default to include proportions of the chi-squared (disabling that by assigning FALSE)
  chisq = TRUE, # chi-squared
  sresid = TRUE) #standardized residuals)
```

```
##      Cell Contents
## |-----|
## |              N |
## |      Expected N |
## |      N / Row Total |
## |      Std Residual |
## |-----|
##
## =====
##               vote.cleaned$employ
## vote.clend$y.vt  Employed full-tm  Employed part-tm  Not employed  Total
## -----
## Register to vote          140           43           178          361
##                          147.8          44.7          168.5
##                          0.388          0.119          0.493      0.406
##                          -0.642         -0.250          0.730
## -----
## Make easy to vot          224           67           237          528
##                          216.2          65.3          246.5
##                          0.424          0.127          0.449      0.594
##                          0.531          0.206         -0.604
## -----
## Total                   364           110           415          889
## =====
##
## Statistics for All Table Factors
##
## Pearson's Chi-squared test
## -----
## Chi^2 = 1.697634      d.f. = 2      p = 0.428
```

```
#checking expected values for ease of vote and sex
```

```
CrossTable(x = vote.cleaned$easy.vote,
  y = vote.cleaned$sex,
  expected = TRUE, # expected value
  prop.c = FALSE, # default to include proportions in the column (disabling that by assigning FALSE)
  prop.t = FALSE, # default to include proportion out of total (disabling that by assigning FALSE)
  prop.chisq = FALSE, # default to include proportions of the chi-squared (disabling that by assigning FALSE)
  chisq = TRUE, # chi-squared
  sresid = TRUE) #standardized residuals)
```

```
##      Cell Contents
## |-----|
## |              N |
## |      Expected N |
## |      N / Row Total |
## |      Std Residual |
## |-----|
##
## =====
##                                vote.cleaned$sex
## vote.cleaned$easy.vote      Male   Female   Total
## -----
## Register to vote             172     189     361
##                               169.7    191.3
##                               0.476    0.524    0.406
##                               0.174   -0.163
## -----
## Make easy to vote            246     282     528
##                               248.3    279.7
##                               0.466    0.534    0.594
##                               -0.143    0.135
## -----
## Total                        418     471     889
## =====
##
## Statistics for All Table Factors
##
## Pearson's Chi-squared test
## -----
## Chi^2 = 0.09570961      d.f. = 1      p = 0.757
##
## Pearson's Chi-squared test with Yates' continuity correction
## -----
## Chi^2 = 0.05805901      d.f. = 1      p = 0.81
```

As the expected values are sufficient for both variables, that is 5 or higher in at least 80% of groups , so the assumption was met.

#Violation 3: Violating the independent observations assumption The data does not appear to have repeated measures (e.g., before/after interventions or siblings in the same dataset). Both sex and employment status are independent variables because they are being used to explain opinions about ease of voting, hence assumptions are met.